

Task 5 - Machine Stress Prediction using Machine Learning

Problem Statement:

In manufacturing industries, machines operate continuously under different loads and temperatures. High stress on machines can lead to breakdowns and production loss.

This project predicts the **Stress Score of a machine** using operational parameters.

The system helps in:

- Detecting high stress conditions
- Suggesting maintenance
- Preventing machine failure

Objective:

The objective of this project is:

- To build a regression model that predicts machine stress score.
- To help industries monitor machine health.
- To classify whether maintenance is required or not.

Dataset Description:

The dataset contains machine operational parameters:

Feature	Description
Type	Machine Type: L = Low, M = Medium, H = High
Air temperature [K]	Surrounding air temperature
Process temperature [K]	Internal process temperature
Rotational speed [rpm]	Speed of machine rotation

Torque [Nm]	Load applied to machine
Stress Score (Target)	Calculated stress value

Feature Engineering:

We created one additional feature: Because temperature difference affects machine stress.

temp_diff = Process temperature - Air temperature

Data Preprocessing:

Steps performed:

1. Dropped unnecessary columns (UDI, Product ID, Failure Type).
2. Handled categorical feature (Machine Type) using One-Hot Encoding:
 - Type_L
 - Type_M
 - Type_H
3. Split data into:
 - 80% Training
 - 20% Testing

Model Used:

Stress Score Regression Model (custom regression model using Random Forest/ensemble technique):

- Target variable: continuous Stress Score
- Regression model chosen for numeric prediction

- Feature selection and temp_diff engineered feature improved performance

Model Evaluation Metrics:

Metric	Value
Mean Absolute Error	4964.47
Root Mean Square Error	8443.86
R-Squared	0.9952
Mean Absolute Percentage Error	0.84%
Accuracy	99.16%

Interpretation:

- Model explains ~99.5% of variance (very high R^2)
- Very low prediction error (MAPE <1%)
- Model is extremely accurate in predicting Stress Score

Stress Level Logic

- Stress Score < 200,000 → Low Stress (Healthy Machine)
- Stress Score 200,000–500,000 → Medium Stress
- Stress Score > 500,000 → High Stress (Maintenance Recommended)

User Interface:

- Python + Flask used for backend
- HTML/CSS used for frontend input interface

- Users can input machine parameters to get real-time stress score and status

Conclusion:

- The regression model predicts machine stress very accurately
- Supports predictive maintenance
- Reduces machine downtime and prevents failures